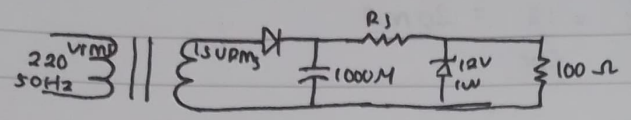
Nama : Ailsa Rahma
NIM : 2411331002

No.
Date

# Dioda zener, BJT dan FET

1. 


a). Besarnya tegangan DC pada kapasitor

$A = V_p = V_{rms} \cdot \sqrt{2}$
$= 15 \cdot \sqrt{2} = 21{,}21\,V$

$V_c = V_p - V_{SI1}$
$= 21{,}21 - 0{,}7 = 20{,}51\,V$

b). Arus max pada dioda zener

$I_{max} = \frac{P_{Zmax}}{V_Z} = \frac{1}{12} = 83{,}3\,mA$

c). Nilai Rs terkecil supaya $P_{Zmax}$ tidak terlampaui

$I_S = \frac{V}{R} = \frac{12}{100} = 0{,}12\,A = 120\,mA$, $V_o = 12V$

$R_S = \frac{V_C - V_O}{I_{Zmax} + I_S} = \frac{20{,}51 - 12}{83{,}3 + 120} =$ ~~0,041~~ 0,041 mΩ

2. Rangkaian stabilizer tegangan pada gambar-2 menggunakan transistor dengan $\beta = 100$, $V_{BE} = 0{,}7V$ dan dioda zener 12V. Bila $V_{in} = 21{,}4V$, $I_Z = 1{,}75mA$ $I_L = 25\,mA$

a). Nilai Rs

- $I_E = I_C + I_B = 25mA$
  $\beta = \frac{I_C}{I_B} \leftrightarrow I_B = \frac{25}{100} = 0{,}25mA$
- $I_S = I_Z + I_B$
  $= 1{,}75 + 0{,}25 = 2mA$
- $R_S = \frac{V_{in} - V_Z}{I_S} = \frac{21{,}4 - 12}{4} = 4{,}7\,k\Omega$

b). Besar tegangan pada RL

$V_L = V_Z - V_{SI1}$
$= 12 - 0{,}7 = 11{,}3V$

c). besar tegangan VCE

$V_{CE} = V_C - V_E$
$= 21{,}4 - 11{,}3 = 10{,}1\,V$

d). $P_{transistor} = V_{CE} \times I_C =$
$8{,}7 \times 0{,}025 = 0{,}2175\,W$

3. Dik : $V_{CC} = 12V$ $V_{CE} = 1/2\,V_{CC} = 6V$
$\beta = 100$ $I_C = 10\,mA$

a). $V_{RC} = V_{CC} - V_{CE} = 6V$
$R_C = \frac{V_{RC}}{I_C} = \frac{6}{10\,mA} = 600\,\Omega$

b). $I_B = \frac{I_C}{\beta} = \frac{10mA}{100} = 0{,}1mA$

$V_{RB} = V_{CC} - V_{BE} - V_{RC}$
$= 12 - 0{,}7 - 6 = 5{,}3V$
$R_B = \frac{V_{RB}}{I_B} = \frac{5{,}3}{0{,}0001} = 53k\Omega$

No.

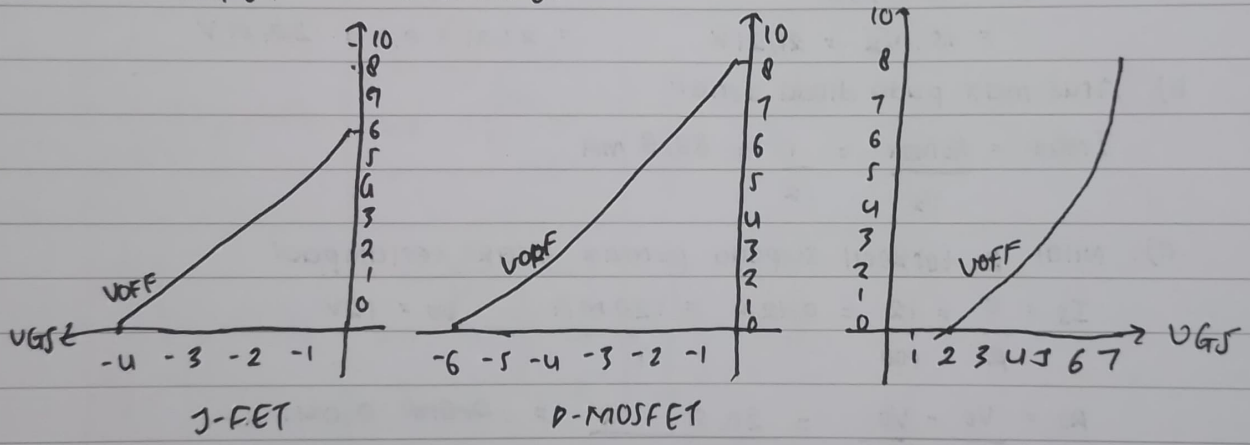
Date

c). garis beban dapat ditentukan 2 titik

ketika $I_C = 0, V_{CE} = V_{CC} = 12V$

ketika $V_{CE} = 0, I_C = \frac{V_{CC}}{R_C} = \frac{12}{600} = 20mA$

4. a). gambarkan kurva karakteristik transkonduktansi $I_D = V_{GS}$ untuk JFET, D-MOSFET dan E-MOSFET




J-FET D-MOSFET

b). bila diinginkan $V_{GSQ} = \frac{1}{4} V_{GSOFF}$ dan $V_{DSQ} = \frac{1}{2} V_{DD}$. tentukan nilai $R_S$ dan $R_D$ bila parameter JFET, ~~tentukan~~ sbb: $I_{DSS} = 8mA$

$V_{GSOFF} = -4V$

↳ $I_D = I_{DSS}\left[\frac{1 - V_{GS}}{V_{OFF}}\right]^2 = 8mA\left[\frac{1 - \frac{-1}{-4}}{}\right]^2 = 4,5mA$

$V_{GS} = V_G - V_S$ $R_S = \frac{V_S}{I_D} = \frac{1k}{4,5} = 222,2\Omega$

$V_G = 1kV$

c). Tentukan besar $R_G$ sehingga input mempunyai tahanan sebesar $1M\Omega$

↳ karena nilai Gate = $\infty$ maka besar $R_G$ jika nilai input $1M\Omega = 1M\Omega$